Flutter Data Storage and Persistence – Lab Sheet (1 to 17)

1. Map – Student Details

Code:

```
var s = {'name':'Puja','roll':101,'course':'BCA'};
print(s);
```

Output:



2. Object → Map

Code:

```
var m = s.toMap();
print(m);
```

Output:



3. Map → JSON

Code:

```
String j = jsonEncode(map);
print(j);
```

Output:



4. Save User Name

Code:

```
p.setString('name','Puja');
print(p.getString('name'));
```

Output:



5. Counter

Code:

```
int n = (p.getInt('count') ?? 0) + 1;
print(n);
```

Output:



6. Dark Mode

Code:

```
p.setBool('dark', true);
print(p.getBool('dark'));
```

Output:



7. Remember Me

Code:

```
p.setBool('remember', true);
print(p.getBool('remember'));
```

Output:



8. Multiple Values

Code:

```
p.setString('name','Puja');
p.setString('email','puja@gmail.com');
p.setInt('age',20);
```

Output:



9. List of Strings

Code:

```
p.setStringList('sub',['Java','Python','Flutter']);
print(p.getStringList('sub'));
```

Output:



10. Clear SharedPreferences

Code:

```
p.clear();
print('Data Cleared');
```

Output:



11. First Launch

Code:

```
bool first = !(p.getBool('first') ?? false);
p.setBool('first', true);
```

Output:



12. Write to File

Code:

```
file.writeAsString('Hello Flutter');
print('Saved');
```

Output:



13. Read from File

Code:

```
print(await file.readAsString());
```

Output:



14. Student JSON

Code:

```
p.setString('student', jsonEncode(s));
print(jsonDecode(p.getString('student')));
```

Output:



15. To-Do List

Code:

```
p.setStringList('todo',['Study','Coding','Assignment']);
print(p.getStringList('todo'));
```

Output:



16. Notes JSON

Code:

```
file.writeAsString(jsonEncode(['Note 1','Note 2']));
print(jsonDecode(await file.readAsString()));
```

Output:



17. Student Profile

Code:

```
p.setString('name','Puja');
p.setString('roll','101');
p.setString('course','BCA');
p.setString('sem','5');
```

Output:

